

Evaluation of the Effect of Momentum on Interaction Forces in a Linked System

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Introduction

Newton's first law describes the behavior of an object if no external forces were to act on it and in so doing provides a baseline for understanding the forces applied. In a linked system, a similar counterfactual analysis can show the behavior if supplied joint torques were not present at a given kinematic state. This type of analysis can provide insight into the nature of the system and how the forces seen are affected by its current momentum state.

The golfer – club system represents a linked system consisting of multiple joints and segments. At each of these joints a torque can be applied by the golfer to drive the club throughout its movement. The interaction force developed between segments depend on both the torques applied at each joint and the kinematic state of the system and the momentum it contains [1].

A forward dynamics model is used as a basis for evaluating the behavior of the system when applied torques at joints are “turned off” in stepwise fashion across a representative number of time points. Output from these runs were compiled creating a representation of the contribution of momentum to joint interaction forces at each point in time. This analysis separates the contribution of “what has already been done” (forces created through interaction of momentum) from “what is currently being done” (forces created by applied joint torques) to generate the total force observed at the joints in a linked system. This is similar to the effect of initial conditions on the solution of differential equations.

Methods

A two hand flexible shaft forward dynamics model was constructed for the golf swing utilizing Matlab Simulink version R2023a. Motion of arms and club were confined to a plane. The model is driven by variable torque inputs at each joint. Trial and error iteration was used to achieve a representative golf swing with clubhead speed and angular velocities on a similar scale as those found in professional swings [2].

Over the course of 56 runs in the model, step changes driving the joint torques to zero were evenly spaced out over the time range of the downswing. For each iteration the force of each hand on the club with no driving torque present was compiled. These forces were compared to the total force applied to the club when joint torques were present.

Results and Discussion

In Figure 1, the results of the analysis are represented as dots with the counterfactual (CF) label and show the momentum contribution to interaction forces generated. The total forces are depicted as solid lines. The axial and normal forces within the plane of the swing are shown relative to the club handle in a local coordinate system.

The contribution of momentum to the forces normal to the club is stable until approximately 0.125 seconds. At this point, the club begins to release and the momentum driven component of the force reverses magnitude prior to the force reversal seen in the total force. The momentum contribution to normal force is higher than the total normal force up until approximately 0.15 seconds into the downswing indicating that there are competing effects influencing force developed.

The contribution of momentum to axial forces is initially small and increases in magnitude as the club releases in late downswing. The contribution of the momentum component to the total force appears to be more significant in the lead hand than it is in the trail hand.

The momentum driven component of force contributes a sizeable portion of the total force and may help explain the high rates of force development seen in the interaction forces. Interaction of momentum between segments at high velocity is not bound by the same constraints as muscularly driven efforts of the golfer such as force-velocity curves, force-length curves, or limitations to rate of force development due to latency. Momentum driven components of force will exhibit rapid changes, but these are inherent to the kinematics of the system and are not reliant on instantaneous activation states of the muscles driving the action.

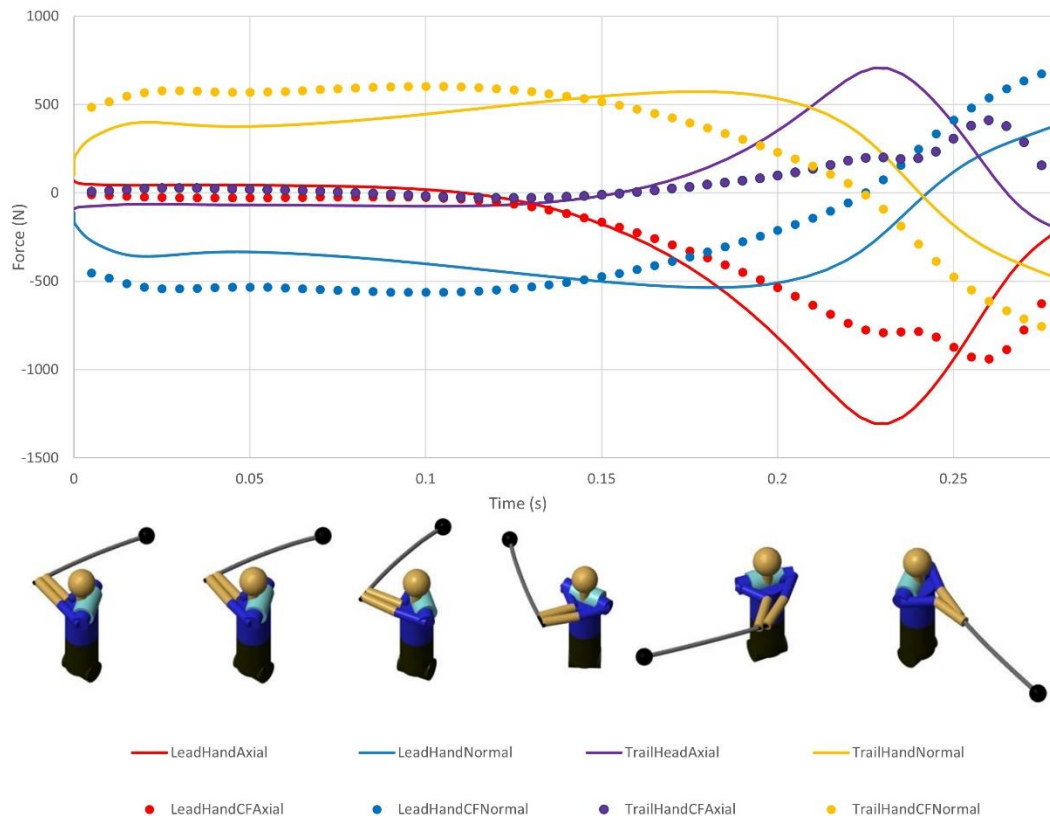


Figure 1 - Contribution of Momentum to Force Applied to Club

References

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